

Natural Language Processing

Introduction

Natural Language Processing (NLP) is an area of AI concerned with enabling computers to process, understand, and generate human language.

Common Tasks

NLP tasks include text classification, sentiment analysis, machine translation, question answering, information extraction, summarization, and named entity recognition.

Text Representation

Traditional NLP systems can represent text using methods such as bag-of-words and TF-IDF. Modern systems often use vector embeddings that represent semantic information in numerical form.

Large Language Models

Large language models are trained on large collections of text and can perform many language tasks. They can generate text, answer questions, summarize content, and support conversational applications.

Challenges

NLP systems can struggle with ambiguity, context, rare language patterns, factual errors, bias, and multilingual differences.